

1C

1c) What is the Big O complexity of

i) $10n^3 + 3n^2 - \underbrace{10n + 4}_{\text{ignoring constants}}$

$$10n^3 + 3n^2$$

$$\underline{\underline{O(n^3)}}$$

ii) $\log(3n^2)$

$$\log(3n^2) = \log 3 + 2 \log n$$

$$= O(\log n)$$

$$\underline{\underline{O(\log n)}}$$

iii) $\frac{n^2 + n}{2}$

$$\frac{n^2 + n}{2} = \frac{n^2}{2} + \frac{n}{2}$$

(Since $\frac{1}{2}$ does not affect Big O)
the dominant is $O(n^2)$